

Descriptor: *The Residential 1 Hz Energy Dataset (R1Hz)*

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ABSTRACT The Residential 1 Hz Energy Dataset (R1Hz) contains circuit-level electrical measurements from one side-attached duplex in Burnaby, British Columbia, Canada. Four aligned 1 Hz files provide current, power factor, real power, and reactive power for one main aggregate and 18 named circuit/load channels over 757 days, from 13 September 2017 through 9 October 2019. Each file contains 65,404,800 rows. The release also contains hourly, daily, and monthly interval-energy products; native in-home-display (IHD) observations; hourly utility energy; outdoor climate observations; a channel dictionary; and the original daily Modbus records, monthly climate downloads, and utility exports retained for provenance. Two acquisition outages affected 0.203758% of the circuit-value cells in the four 1 Hz streams. Missing circuit values were imputed deterministically from weekday- and local-clock-aligned observations 364 days away; 13 additional isolated real/reactive-power seconds were estimated from contemporaneous electrical measurements. Every row containing a synthetic value is marked *s*. This paper documents the instrumentation, exact released schemas, recovery methods, validation, remaining limitations, and storage organization needed for reproducible use of R1Hz.

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DATA TYPE/LOCATION Time series; Burnaby, British Columbia, Canada

INDEX TERMS energy data, imputation, load disaggregation, non-intrusive load monitoring, power measurement, residential electricity, smart meter

BACKGROUND

Public, high-resolution residential electricity datasets are important for evaluating load forecasting, anomaly detection, demand response, energy feedback, and non-intrusive load monitoring (NILM) under realistic operating conditions. Their value depends not only on sampling frequency, but also on an explicit time base, named loads, stable schemas, provenance, quality flags, and disclosure of any synthetic values.

R1Hz complements datasets that trade temporal resolution, monitoring duration, and number of sites in different ways. SmartNIALMeter includes many buildings and appliances at approximately 5 s resolution [1]; Plegma combines appliance and contextual observations in Greece [2]; and

RAE provides an earlier Canadian residential smart-meter resource [3]. Reviews of reproducible NILM research emphasize that transparent preprocessing and consistent evaluation are as important as open access [4]. R1Hz contributes a continuous 1 Hz index, 19 named electrical channels, multiple electrical quantities, interval-energy derivatives, and retained native acquisition records for a single deeply monitored residence. Its purpose is detailed within-site analysis rather than population inference.

COLLECTION METHODS AND DESIGN

A. Monitoring Site and Instrumentation

The monitored site is one side-attached duplex in Burnaby, British Columbia, in the America/Vancouver time zone.

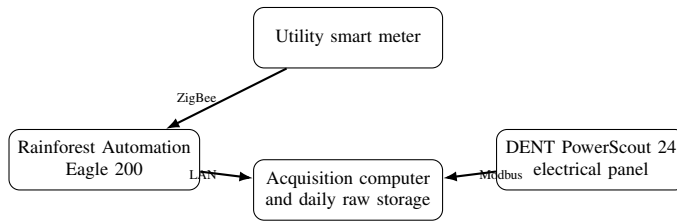


FIG. 1. Collection architecture. The Eagle 200 receives utility smart-meter data, the PowerScout 24 measures branch circuits, and the acquisition computer stores the native records.

The dwelling was built in 2016, has two levels (approximately 105.4 and 67.5 m²), and had approximately three occupants during collection. These descriptors provide context only; no occupant-level variables are released.

Branch-circuit measurements were collected with a DENT PowerScout 24 installed at the electrical panel. Twenty-one physical meter inputs were converted into 19 logical channels: *main* combines the two service legs on meters 1 and 2, and *dryr* combines the two dryer legs on meters 4 and 5. The other 17 logical channels each map to one physical input. Current, power factor, real power, and reactive power were placed on a common 1 Hz index. A Rainforest Automation Eagle 200 ZigBee gateway/in-home display (IHD) supplied native smart-meter power and cumulative-energy observations at an irregular interval of approximately 8–15 s. The release contains 1,619,689 native IHD records from 13 September 2017 through 13 September 2018. Separate hourly utility-energy records cover 9 June 2016 through 20 November 2019, and hourly outdoor climate records cover 1 September 2017 through 31 October 2019. The utility records were exported from the principal researcher’s own residential BC Hydro account; the researcher, homeowner, and account holder are the same person.

B. Outdoor Climate Source

Hourly meteorological observations were obtained from Environment and Climate Change Canada’s Historical Climate Data service for *VANCOUVER INTL A*, British Columbia (Climate Identifier 1108395; WMO Identifier 71892; Transport Canada Identifier YVR) [5]. The station is listed at 49.19° N, 123.18° W and 4.30 m elevation; its current operator is NAV Canada. The word “current” does not imply that NAV Canada operated the station throughout the complete historical record.

The downloaded source timestamps use local standard time (PST) throughout the year. Following the source instruction, one hour was added where daylight-saving time was observed before representing the data in *America/Vancouver* local civil time. The released fields and units are: air temperature and dew point in °C; relative humidity in percent; precipitation amount in mm; wind direction in tens of degrees true (the direction from which wind blows, with 0 denoting calm); wind speed in km/h;

visibility in km; and station pressure in kPa. Humidex and wind chill are indices, and *weather* is categorical text [6].

Source quality flags are not copied into separate released columns. In the source, E denotes estimated, M missing, NA not available, and a blank denotes an unobserved value; D, where present, denotes data subject to further quality control. For a NAV Canada station’s *Weather* field specifically, the source FAQ explains that NA means no special weather elements were reported rather than an unavailable quantitative observation [7]. Redistribution follows the source site’s Limited Use Software and Data Product Licence Agreement. It permits redistribution without an explicit fee for the ECCC product provided that ECCC is acknowledged and recipients accept the same redistribution restrictions; the repository’s software licence does not govern the climate observations [8].

Release-format validation found 402 of the 18,984 processed climate rows (2.1176%) parsing as 16 or 17 columns because a compound *weather* value contained one or two unquoted commas. The repair enclosed only the complete final field on those rows in CSV double quotes. All timestamps and parsed values are identical before and after this serialization repair, so these rows are not synthetic and are not marked *s*. Five raw ECCC records contain station/time fields but no observation fields; their timestamps are retained in *climate.csv* with blank observation fields.

C. Utility Export Source

The source utility records are two original, non-overlapping BC Hydro account exports containing 30,240 hourly interval-start rows in total. Their nine native fields are account holder, account number, interval start, net consumption (kWh), demand (kW), power factor (%), estimated-usage flag, service address, and city. The retained exports suppress the account-holder and address fields; the full account number is absent. Net consumption contains 172 N/A values (0.568783%), and 48 numeric intervals carry the source estimated-usage flag. Demand and power factor are unavailable throughout. The source timestamps have no UTC offset: repeated fall-back labels occur in 2016–2018, while the 2019 export contains only one 01:00 row on the fall-back day. The released *utility.csv* preserves the source uncertainty through blank energy values and explicit local-time markers rather than treating naive local time as a unique key.

D. Pre-processing and Aggregation

The native Modbus records are retained unchanged in *raw_modbus/*. The processed circuit streams use lower-case logical channel identifiers from *appliances.csv*. Apart from the sparse native IHD field in *power.csv*, the four 1 Hz files have the same 65,404,800 consecutive timestamps and the same 19 circuit channels. Real power and reactive power are stored as integer watts and var, current in amperes, and power factor as a dimensionless ratio.

The three interval-energy products use watt-hours. An hourly record labelled at time T summarizes the preceding hour $[T - 1\text{h}, T)$. Daily records group hourly labels by local date, and monthly records group daily labels by local year-month. Consequently, the first monthly record begins on 13 September 2017 and the final monthly record is a partial October 2019. Users requiring strict civil-day energy boundaries should account for this label-grouping convention and daylight-saving transitions.

E. Filling Missing Data

The original four circuit streams already contained a complete, ordered 1 Hz timestamp index: there were no absent, duplicated, reversed, or malformed timestamp rows. The missing data were blank circuit measurement fields within that index. Two outages were common to all four quantities: 8 June 2018 11:21:53 through 9 June 2018 19:35:34 PDT (116,022 s), and 1 June 2019 02:32:34 through 07:19:52 PDT (17,239 s). Thirteen additional seconds contained blanks only in real and reactive power.

The long outages were filled by deterministic, same-channel donor substitution at a 364-day offset. A 364-day shift preserves weekday and local clock time: the June 2018 outage uses 7–8 June 2019, while the June 2019 outage uses 2 June 2018. Donor intervals were checked for null circuit fields. Target timestamps and any originally observed values were retained; for example, one measured `current/gen2` value inside the first outage was not overwritten. This procedure is pattern-based imputation, not stochastic simulation, and the resulting values are plausible surrogates rather than recovered ground truth.

For the 13 isolated seconds, channel-specific real power was estimated from nearby valid observations using

$$\hat{P}_{c,t} = \hat{\alpha}_c + \hat{\beta}_c(I_{c,t} PF_{c,t}), \quad (1)$$

then rounded to whole watts. Reactive power was linearly interpolated between the nearest valid bracketing observations for each channel and rounded to whole var. These methods use contemporaneous current and power factor or immediate temporal neighbours instead of an opposite-year donor when the gap is isolated.

Hourly circuit-energy gaps were reconstructed from recovered 1 Hz real power. Of 135 reconstructed utility-hour cells, 90 used the mean of at least 440 native IHD samples in the associated source hour, rounded to the utility series' 10 Wh resolution; 14 used a month-local linear model of utility energy on main energy with partial IHD available as a check; and 31 used that model without IHD coverage. Daily and monthly products were then rebuilt from the hourly product so that blanks were not silently skipped. Table 1 gives the exact scope.

VALIDATION AND QUALITY

Full-file scans verified the row count, schema, timestamp order, and null count of each recovered 1 Hz stream. All

originally observed circuit measurements were compared with their source counterparts and were unchanged; only targeted blank cells were populated. The recovered streams contain no remaining blank circuit measurements. The sparse `ihd` column in `power.csv` is expected because it is populated only when a native IHD observation exists.

Aggregation checks compared clean observed intervals with values independently recomputed from the higher-frequency source. For 3,192 observed circuit-hour comparisons, aggregation from 1 Hz real power had a mean absolute error (MAE) of 0.263 Wh and a 95th-percentile absolute error of 1 Wh. Direct IHD aggregation had an MAE of 1.77 Wh over 3,379 observed utility hours (95th percentile 10 Wh). The month-local utility-to-main comparison had an MAE of 1.181 Wh over 9,364 observed hours (95th percentile 10 Wh). Hourly-to-daily and daily-to-monthly reconciliation after recovery both had a maximum absolute difference of 0 Wh.

Quality flags and remaining limitations must be retained in downstream analysis. The 19 blank circuit cells at the first hourly record are intentional because the preceding hour lies outside 1 Hz coverage. The source `utility.csv` contains 172 blank hourly-energy values among 30,240 rows (0.568783%); the two original exports additionally identify 48 numeric intervals as estimated. The native `ihd.csv` includes 76 retained duplicate timestamps. Five climate hours have all observation fields blank; `precip_amt` is blank throughout the current climate file, while `humidex` and `windchill` fields are conditionally applicable. The pre-existing `+` marker denotes a row added during the original time-series regularization; it occurs on 64 `current/power-factor` rows and 51 `real/reactive-power` rows and is distinct from the later recovery marker `s`. Synthetic intervals should normally be excluded from event-timing, transient, and ground-truth NILM evaluation, or included only in a clearly reported sensitivity analysis.

RECORDS AND STORAGE

The authoritative dataset record is <https://doi.org/10.7910/DVN/RCB5VJ> [9]. Table 2 lists the logical release files. Counts exclude the header row. Sizes are uncompressed sizes of the canonical recovered files; compressed archive sizes will differ.

The four uncompressed 1 Hz CSVs exceed the Harvard Dataverse 2.5 GB per-file limit. They may therefore be distributed as compressed archives or as date-partitioned compressed members. The raw files may likewise be grouped into documented archives. Archive member names should preserve the logical names in Table 2; a release manifest should record each member's date range, uncompressed byte count, row count, and checksum. If a compressed member still exceeds the repository limit, partitioning by documented contiguous date ranges is preferable to opaque multipart archives.

TABLE 1. Missing-data recovery in the release. Percentages for the four 1 Hz files use 65,404,800 timestamp rows or 65,404,800 × 19 circuit cells per file.

File	Rows marked s	Imputed cells	Cells in scope	Imputed (%)	Method / remaining blanks
current.csv	133,261	2,531,958	1,242,691,200	0.203748	364-day, same-channel donor; none remain
power_factor.csv	133,261	2,531,959	1,242,691,200	0.203748	364-day, same-channel donor; none remain
power.csv	133,274	2,532,206	1,242,691,200	0.203768	Donor plus local regression; none remain in circuit fields
reactive.csv	133,274	2,532,206	1,242,691,200	0.203768	Donor plus local interpolation; none remain
Four-file total	—	10,128,329	4,970,764,800	0.203758	Sparse ihd field excluded from circuit-cell denominator
energy_hourly.csv	174	876	363,360	0.241083	741 circuit and 135 utility cells; 19 first-record circuit blanks remain
energy_daily.csv	47	—	—	—	Rebuilt from hourly values; no energy blanks remain
energy_monthly.csv	21	—	—	—	Rebuilt from daily values; no energy blanks remain

TABLE 2. Release inventory. “1 Hz” means one row per second; the IHD cadence is native and irregular. Raw-folder counts exclude operating-system and version-control placeholder files.

Logical path	Grain	Rows / files	Coverage (local time)	Size	Description
appliances.csv	dictionary	19	not applicable	895 B	Logical-channel names, physical meter mapping, and mixed-load notes
climate.csv	hourly	18,984	2017-09-01–2019-10-31	1.272 MiB	Outdoor temperature, humidity, wind, pressure, visibility, and condition fields
current.csv	1 Hz	65,404,800	2017-09-13–2019-10-09	6.581 GiB	Circuit current (A)
power_factor.csv	1 Hz	65,404,800	2017-09-13–2019-10-09	7.736 GiB	Circuit power factor (dimensionless)
power.csv	1 Hz	65,404,800	2017-09-13–2019-10-09	4.768 GiB	Circuit real power (W) and sparse aligned IHD power
reactive.csv	1 Hz	65,404,800	2017-09-13–2019-10-09	4.604 GiB	Circuit reactive power (var)
energy_hourly.csv	hourly	18,168	2017-09-13–2019-10-09	1.457 MiB	Utility, main, and circuit interval energy (Wh)
energy_daily.csv	daily	757	2017-09-13–2019-10-09	81.546 KiB	Daily-labelled energy aggregates (Wh)
energy_monthly.csv	monthly	26	2017-09–2019-10	3.724 KiB	Monthly-labelled energy aggregates (Wh); edge months are partial
ihd.csv	native 8–15 s	1,619,689	2017-09-13–2018-09-13	67.954 MiB	Native IHD power (W) and cumulative meter energy (kWh)
utility.csv	hourly	30,240	2016-06-09–2019-11-20	1.039 MiB	Native utility interval energy (Wh, 10 Wh resolution)
raw_modbus/SUB_YYYY-MM-DD.csv	one local day/file	typically 691,200/file	2017-09-13–2019-10-09	76.711 GiB total	Headerless, immutable device-native Modbus register records; 757 files
raw_climate/	monthly source files	18,984 / 31 files	2017-09-01–2019-10-31	4.587 MiB total	26 ECCC hourly CSVs, station inventory, metadata, and historical helper script
raw_utility/	account exports	30,240 / 2 files	2016-06-09–2019-11-20	1.833 MiB total	Immutable BC Hydro exports with source flags and suppressed direct identifiers

Three raw-source directories form the provenance and audit layer. `raw_modbus/` contains one file for every local date in the 757-day acquisition interval, totalling 82,367,715,595 bytes (76.711 GiB). Each headerless `SUB_YYYY-MM-DD.csv` contains 45 fields: a Unix timestamp, a register-bank identifier A–H, and 43 raw integer values corresponding in order to Modbus holding registers 4021–4063. A complete ordinary day has eight bank rows for each of 86,400 seconds, or 691,200 rows. Counts can differ when a day is partial or daylight-saving time changes; for example, the partial 9 June 2018 file has 126,912 rows and the 4 November 2018 fall-back file has 720,000 rows.

`raw_climate/` contains the 26 original monthly ECCC downloads, the station-inventory snapshot and supporting metadata, and a historical concatenate-only helper. The five short source records remain unchanged for provenance. The helper is retained for history but was not used to create the release because it leaves repeated headers and embedded byte-order marks. `raw_utility/` contains two original BC Hydro exports with contiguous, non-overlapping date coverage. Direct account-holder, account-number, address, and city values are suppressed in their data rows, although the pseudonymized filenames retain four account digits. These raw files are not imputed or normalized. The root-level files are canonical processed products. Retaining both

TABLE 3. Marker values used in released CSVs.

Marker	Meaning
blank	Ordinary record with no special provenance flag
+	Row added during the original 1 Hz time-series regularization. This legacy flag is retained separately from later imputation.
s	At least one value in the row is synthetic/imputed. In daily or monthly energy, at least one lower-grain input is synthetic.
d	Retained duplicate native timestamp in <code>ihd.csv</code>
t	Local-time discontinuity associated with a daylight-saving-time transition in hourly utility/climate series

layers allows users to audit register conversion, channel mapping, source flags, timestamp treatment, and recovery without mistaking synthetic values for device observations.

F. Temporal Fields and Markers

`unix_ts` is an integer Unix timestamp in seconds with UTC as its reference. `date/time` and `local_dt/local_tm` are local civil-time representations in America/Vancouver. The `span` field is H, D, or M for hourly, daily, or monthly aggregates. Daylight-saving placeholders and retained native duplicates mean that `unix_ts` is not globally unique in every auxiliary file; users should interpret it together with `marker`. Table 3 documents all marker values present in the release.

G. Logical Electrical Channels

The channel dictionary in Table 4 replaces generic and incorrect `sub?` labels. A “mixed” channel can contain more than one end use and is not appliance-level ground truth.

H. Exact CSV Schemas

Table 5 gives the ordered fields rather than unsupported generic `sub1-sub24` columns. All processed electrical identifiers are lowercase. For compactness, $\mathcal{C}$ denotes the 19 ID/CSV fields in Table 4, in the exact top-to-bottom order shown there.

INSIGHTS AND NOTES

I. Energy Composition

The recovered monthly product totals 6,804.210 kWh at the utility meter and 6,671.084 kWh at the monitored house sub-panel. The 18 non-main logical channels sum to 6,547.487 kWh. These quantities are related but not interchangeable: the utility meter includes a garage load that is outside the monitored sub-panel, and circuit sums can differ from `main` because of metering coverage, instrument accuracy, rounding, and synthetic recovery. Figure 2 therefore excludes both utility and `main` from the component composition and reports them separately in the text.

J. Recommended Use

R1Hz supports circuit-load characterization, short-horizon forecasting, cross-meter comparison, missing-data research, and NILM experiments in which `main` is the aggregate and selected non-mixed circuits provide submeters. It does

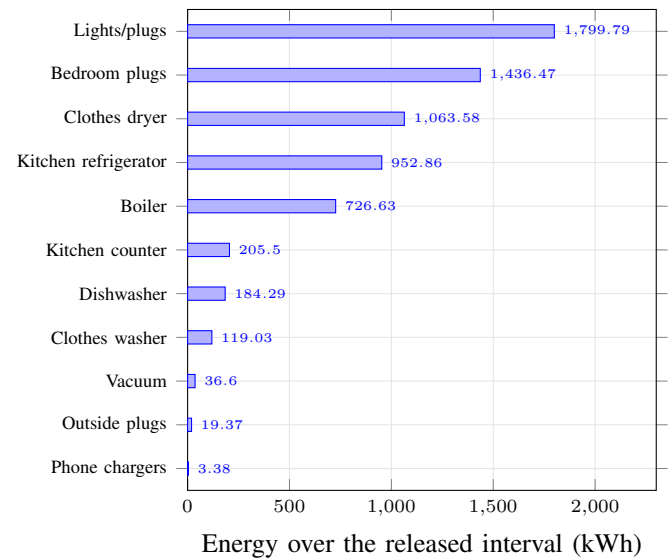


FIG. 2. Non-overlapping component energy from the 18 non-main logical channels, grouped by description. Values are kWh. The utility and main aggregates are deliberately excluded to avoid double counting downstream circuits.

not provide appliance state labels, occupancy labels, or a representative sample of Canadian households. Mixed-load circuits should not be treated as single appliances. Researchers should publish the exact time range, channel set, treatment of daylight-saving time, and treatment of `s-`marked rows used in each experiment.

The native `ihd.csv`, aligned sparse `ihd` field in `power.csv`, and hourly `utility.csv` have different sampling semantics. They should not be filled or resampled implicitly. Similarly, recovered long-gap values preserve daily and weekly timing patterns but cannot reproduce the actual appliance events that occurred during the outages.

SOURCE CODE AND SCRIPTS

The versioned collection, processing, recovery, and validation code is maintained at github.com/smakonin/R1Hz.dataset. The repository includes the deterministic recovery scripts, the climate serialization repair, method inputs for the isolated estimates, a cell-level recovery log, validation summaries, SQL schemas, and the manuscript source. The Harvard Dataverse DOI remains the canonical citation for the data; the Git commit used for a study should additionally be recorded to identify the exact processing implementation.

ETHICS STATEMENT

The principal dataset researcher, S.M., is also the homeowner and BC Hydro account holder. S.M. authorized collection and public release; no separate property-owner or account-holder authorization was required. The measurements originate from a private residence and can reveal patterns of household activity even though no names, full utility account identifiers, audio, images, or exact street address are released.

TABLE 4. Processed logical channel dictionary. Meter numbers refer to PowerScout physical inputs.

ID / CSV field	Meter input(s)	Description	Mixed	Notes
main	1+2	House sub-panel	Y	Utility also supplies an unmetered garage load outside this sub-panel
beda	13	Bedroom plugs	Y	AFCI arc-fault plugs
bedp	6	Bedroom plugs	Y	—
boil	8	Boiler	N	Hot water and radiant heating
chrg	21	Phone chargers	N	Garburator and microwave were not installed
cwsh	10	Clothes washer	N	—
dryr	4+5	Clothes dryer	N	Two electrical legs combined
dwsh	19	Dishwasher	N	—
frdg	11	Kitchen refrigerator	N	—
gen1	3	Lights and plugs	Y	General circuit
gen2	9	Lights and plugs	Y	General circuit
gen3	12	Lights and plugs	Y	Includes Internet/network equipment
gen4	16	Lights and plugs	Y	General circuit
gen5	17	Lights and plugs	Y	General circuit
gen6	20	Lights and plugs	Y	General circuit
kit1	14	Kitchen counter plugs	Y	—
kit2	15	Kitchen counter plugs	Y	—
outp	18	Outside plugs	Y	—
vacu	7	Built-in vacuum	N	—

TABLE 5. Ordered schemas of the canonical processed CSVs and retained raw source CSVs. Here *C* is the ordered 19-channel sequence defined above.

File(s)	Ordered columns
appliances.csv	appliance_id, name, meter_l1, meter_l2, is_mains, has_mixed_loads, notes
climate.csv	unix_ts, marker, local_dt, local_tm, temp, dew_point, rel_hum, precip_amt, wind_dir, wind_spd, visibility, stn_press, hmdx, wind_chill, weather
current.csv, power_factor.csv, reactive.csv	unix_ts, marker, date, time, followed by <i>C</i>
power.csv	unix_ts, marker, date, time, ihd, followed by <i>C</i>
energy_hourly.csv, energy_daily.csv, energy_monthly.csv	unix_ts, span, marker, local_dt, local_tm, utility, followed by <i>C</i>
ihd.csv	unix_ts, marker, date, time, power, energy
utility.csv	unix_ts, marker, local_dt, local_tm, energy
raw_modbus/SUB_YYYY-MM-DD.csv	Headerless: unix_ts, bank identifier A–H, then 43 integer register values for addresses 4021–4063 in ascending order
raw_climate/en_climate_hourly_*.csv	Native ECCC 30-column order: longitude, latitude, station name, climate ID, date/time (LST), year, month, day, and time (LST); then temperature, dew point, relative humidity, precipitation, wind direction, wind speed, visibility, station pressure, humidex, and wind chill, each followed by its source flag; then weather
raw_climate/Station_Inventory_EN.csv	Three metadata lines, then: Name, Province, Climate ID, Station ID, WMO ID, TC ID, Latitude (Decimal Degrees), Longitude (Decimal Degrees), Latitude, Longitude, Elevation (m), First Year, Last Year, HLY First Year, HLY Last Year, DLY First Year, DLY Last Year, MLY First Year, MLY Last Year
raw_utility/*.csv	Account Holder, Account Number, Interval Start Date/Time, Net Consumption (kWh), Demand (kW), Power Factor (%), Estimated Usage, Service Address, City

Location is reported only at city level. Users should not attempt to re-identify residents or infer sensitive household behaviour beyond the scientific purpose of their study.

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Author contributions: S.M. curated the data. D.J.A. and S.M. analyzed the data. D.J.A., S.M., and P.P. reviewed the content and contributed to the manuscript. All authors reviewed the manuscript.

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